



DIBBs eCR Viewer

User Acceptance Testing Guide – eCR Coordinator

Created Date: August 6, 2025

Last Updated: April 24, 2026



Prepared by:

U.S. Department of Health and Human Services



**U.S. DEPARTMENT OF
HEALTH AND HUMAN SERVICES**
CENTERS FOR DISEASE
CONTROL AND PREVENTION



Version History

Version	Author	Reviewed By	Revision Date	Revision Notes
1.0	Elizabeth Chinelo Ikejimba	Kristie Clarke	09/17/2025	Revised to contain details for both NBS-integrated Viewer and Standalone eCR Library
1.1	Matthew Goldberg	Kristie Clarke	12/10/2025	Revised based on Kristie’s feedback
1.2	Matthew Goldberg	Michael Wodajo	04/24/2026	Revised to update based on current process



Table of Contents

1	Objectives	3
2	Testing Scope	4
3	Test Plan.....	4
3.1	Prerequisites	4
3.2	Identify eCRs for Data Quality Checks (eCR Coordinator)	5
3.3	Process eCRs (Informatics UAT).....	5
3.4	Feature & Data Quality Testing	6
4	Final Steps of End User Testing (eCR Coordinator).....	8
4.1	SQL Server Testing (SMEs)	8
5	Wrapping Up.....	8
6	Glossary.....	9



1 Objectives

Note: For definitions of any unfamiliar terms or acronyms (e.g., eCR), refer to the Glossary at the end of this document.

This document lays out steps and resources for eCR coordinators to manage user acceptance testing (UAT) for the eCR Viewer in their jurisdictions.

In this guide, we refer to two versions of the product:

1. Integrated eCR Viewer
2. Standalone eCR Library

Following this guidance helps ensure either version of the eCR Viewer is behaving as expected and enables users to identify any bugs that need to be addressed.

This guide contains reference to additional materials outlined below:

- eCR Viewer User Acceptance Testing Guide – End Users
- eCR Viewer User Acceptance Testing Checklist – End Users
- eCR Viewer User Acceptance Testing Checklist – Informatics

2 Testing Scope

UAT for the eCR Viewer and eCR Library requires testing different components of the application:

Component	Relevant UAT section
DIBBs pipeline	“Process eCRs”
Features of the eCR Library	“Feature & Data Quality Testing”
Data quality of the Viewer and eCR Library	“Feature & Data Quality Testing”
Database integration	“SQL Server Testing”

You’ll need to coordinate with staff across your jurisdiction to complete this testing. Typically:

- Informatics staff handle “Process eCRs.”
- Subject matter experts (SMEs), epidemiologists, and case investigators handle “Feature & Data Quality Testing.”
- Program area epidemiologists handle “SQL Server Testing.”

3 Test Plan

3.1 Prerequisites

To enter the UAT phase, it’s assumed that you:

1. Have DIBBs infrastructure set up and available



2. Completed [technical acceptance testing](#) on your DIBBs installation
3. Have verified the connection between your DIBBs installation and your database (note: database only relevant for jurisdictions using the eCR Library)
4. Have sent test eCRs to the Viewer or eCR Library (e.g., Rhapsody)
5. Have network access to the eCR Viewer or eCR Library

If any of these steps are missing, please contact the DIBBs team or your internal IT staff.

3.2 Identify eCRs for Data Quality Checks (eCR Coordinator)

You'll need to identify a minimum of 15 eCRs to use for UAT. To protect patient privacy, these files must contain no identifying data. This can be achieved in two ways, either through synthetic data or anonymized data using the script provided by our team. Please note, for every eCR/RR pair, you must retain both the XML and HTML files, as both formats are required for testing.

If you want to use more eCRs in testing, please do — 15 files is a minimum, not a maximum.

3.3 Process eCRs (Informatics UAT)

Before eCR data can be seen in the eCR Viewer or the eCR Library, it must be processed through the DIBBs pipeline. This processing converts the data to FHIR format and saves it in cloud blob storage (and parses some data into the database for the eCR Library, if applicable).

You'll likely need to work with your jurisdiction's informatician to conduct this portion of UAT. You can use the Informatics UAT Checklist for this portion of testing.

For each eCR in the test set, please:

1. Find both the eICR and RR XML for a given test eCR document.
2. If needed, rename these files `CDA_eICR.xml` and `CDA_RR.xml`, and compress them into a single zip file.
 - 2.1. The eCR Viewer and the eCR Library expect all files to be in this format (it's the default file format from APHL/AIMs).
3. Using your preferred tool (e.g., Rhapsody) send this zip file through the eCR Viewer's or the eCR Library's `/process-ecr` endpoint. Refer to the [API documentation](#) for specific instructions.
 - 3.1. Refer to our [Rhapsody examples](#) for further guidance. Please note that the example communication points only accept the `CDA_eICR.xml` and `CDA_RR.xml` files within the input ZIP. You may need to include an additional step to remove any extraneous files (such as HTML files) before processing.
4. Note the response received on the Informatics UAT Checklist, including any error messages. If there are any errors, check your cloud provider's container logs for any details about each error.



Once this step is finished, you're ready to move onto "Feature & Data Quality Testing."

3.4 Feature & Data Quality Testing

Here, you'll work with the eventual end users (e.g., SMEs, epidemiologists, and case investigators) of the eCR Viewer and the eCR Library (if applicable) to test both the feature functionality and data quality.

You can directly share the eCR Viewer User Acceptance Testing Guide – End Users and accompanying checklist with them, along with your sample eCRs.

For convenience, the same instructions provided to end users in that guide are reproduced below.

3.4.1 Testing Flow

Use the End Users UAT Checklist to confirm completion of specific tasks for each of the steps below. Make sure you pull it up for reference.

You'll complete testing in three steps:

1. Test the general functionality of the eCR Viewer and eCR Library (if applicable).
2. Check the feature functionality of the individual eCR document view.
3. Assess the data quality of each assigned test eCR.

3.4.2 Library Feature Testing

If you only have the NBS-integrated eCR Viewer, then skip to "3.4.3 Individual Record Feature Testing".

To test the feature functionality of the eCR Library, open the End Users UAT Checklist.

Navigate from the "Full Feature Set" tab to open the "Feature Functionality" tab.

1. Open the eCR Library.
2. Verify the following by marking Yes or No:
 - a. Filtering by condition only shows eCRs with that reportable condition present
 - b. Filtering by date only shows eCRs within the specified date range
 - c. Searching by patient name only shows patients who match the search query
 - d. Pagination (moving from page 1-2, changing number of eCRs displayed, etc.) works as expected
 - e. Sorting by name and dates works as expected
 - f. Where applicable, eCRs are grouped by version

3.4.3 Individual Record Feature Testing

Next, we'll check feature functionality when viewing an individual eCR document. You should still be on the "Feature Functionality" tab of your checklist.

1. Open an eCR — any eCR will work here.



- a. From the eCR Library (if you have access to it), click the patient name from the table.
 - b. From NBS/EpiTrax (if you have access to it), click the View eCR button.
2. Verify the following by marking Yes or No:
 - a. The patient banner at the top of the page displays the patient's name and date of birth, and is sticky as you scroll the page
 - b. Side navigation is functional — clicking a tab navigates you to that section of the page
 - c. Accordions on the page expand and collapse as expected
 - d. The Unavailable Information section is present at the bottom of the document

3.4.4 Data Quality Testing

Finally, we'll check data quality of each eCR by comparing each field in the original eCR/RR to the data display in the eCR Viewer and eCR Library and identifying any disparities.

NBS-integrated eCR Viewer

1. Open the “Data Quality” tab of the End Users UAT Checklist for this test. Open an original eCR in HTML format and note the patient name and eCR document ID number.
2. Find that same patient and document in NBS and open the eCR document.
3. To verify the document IDs match, navigate to eCR Metadata > eICR Details > eICR ID within the Viewer and compare it against the document ID from the original eCR.
4. Using the original eCR as your source of truth, check to ensure every data element is also present in the Viewer's eCR document. Consult the End Users UAT Checklist for specific fields. You'll want to check if:
 - a. Any fields are missing entirely from the eCR document
 - b. Any fields are missing a portion of data from the eCR document (e.g., emergency contact name is present but is missing phone number)
 - c. Any fields may be formatted unexpectedly or incorrectly (especially tables)
5. Repeat for every eCR your coordinator has shared.

Standalone eCR Library

1. Open the “Data Quality” tab of the End Users UAT Checklist for this test.
 - a. Open an original eCR in HTML format (or if your jurisdiction doesn't have access to HTML, use the XML) and note the patient name and eCR document ID number.
2. Find that same patient and document in the eCR Library. Check that all expected fields are present and open the eCR document.
3. To verify the document IDs match, navigate to eCR Metadata > eICR Details > eICR ID within the Viewer and compare it against the document ID from the original eCR.
4. Using the original eCR as your source of truth, check to ensure every data element is also present in the eCR document within the eCR Library. Consult the checklist for specific fields. You'll want to check if:
 - a. Any fields are missing entirely from the eCR document
 - b. Any fields are missing a portion of data from the eCR document (e.g., emergency contact name is present but is missing phone number)
 - c. Any fields may be formatted unexpectedly or incorrectly (especially tables)
5. Repeat for every eCR your coordinator has shared.



More Information about Data Quality

The original eCR documents (i.e., the HTML files) are views of the raw eICR, formatted as XML C-CDA.

The eCR Viewer and eCR Library take this same raw file and converts it to a different standard for internal use.

As a result, some data that appears in the eCR Viewer isn't in the original HTML, and some data visible in the HTML isn't shown in the eCR document found within the Viewer and Library.

4 Final Steps of End User Testing (eCR Coordinator)

Once your end users have finished their testing, please share the results directly with the DIBBs team. We'll work with you to prioritize the most important issues to address and retrieve any test data that may be needed.

4.1 SQL Server Testing (SMEs)

If you only have the NBS-integrated eCR Viewer, then skip to "5 Wrapping Up".

The final stage of testing the eCR Library is to evaluate the extended data available in your SQLServer. How to use these tables will be up to you, so please conduct additional testing or analysis as needed. The steps below are only a rough outline.

If eCRs appear in the eCR Library, they should also be available in your database tables. However, you will want to ensure that:

- Each eCR successfully processed through the DIBBs pipeline has a corresponding entry in the eCR table
- Data shown in the eCR Library for an entry is also present in the tables
- Table data is formatted as expected and can be used in SQL queries

You're also welcome to do any additional analysis on this data (e.g., determining what percentage of the time specific fields (columns) are actually populated). If there are any issues with the data in the table, please reach out to the DIBBs team.

5 Wrapping Up

Thank you so much for your assistance in the UAT process! We truly appreciate your help making the eCR Viewer and the Standalone eCR Library as robust as possible. If you have any questions, please don't hesitate to reach out to us at dibbs@cdc.gov.

6 Glossary

API: Application programming interface (API) is a set of functions and procedures that allow the creation of applications that access features or data of another operating system, application, or other service.



eCR: electronic case reporting (eCR) is the automated, real-time exchange of case report information between electronic health records (EHRs) and public health agencies.

eICR: electronic initial case report (eICR) sent from a clinical care provider to determine if the condition code is reportable for its jurisdiction.

EHR: electronic health records (EHR) are the electronic version of a patient’s medical history that providers maintain over time and may include all of the key administrative clinical data relevant to that patient’s care.

eRSD: electronic reporting and surveillance distribution service (eRSD) is a service provided by APHL that looks at the EHR to determine if there’s a trigger code present by using LOINC codes.

LOINC: Logical observation identifiers names and codes (LOINC) is a standardized coding system used for identifying health measurements, observations, and services. LOINC codes facilitate the exchange and aggregation of clinical laboratory data.

NBS: NEDSS Base System (NBS) is a CDC-developed integrated information system that helps STLT public health departments manage reportable disease data and send notifiable disease data to the CDC.

NEDSS: National Electronic Disease Surveillance System (NEDSS) is a set of architectural standards to support NNDSS.

RCKMS: Reportable Conditions Knowledge Management System (RCKMS) is an authoritative, real-time, digital portable to enhance disease surveillance by providing comprehensive information on public health reporting requirements to public health reporters such as clinicians and laboratories.

RR: Reportability response (RR) is a document, sent from a jurisdiction to a clinical care provider, that communicates information on the status of a reportable condition/

Schema: A schema is how a database categorizes recorded data.

STLT: State, tribal, local, and territory (STLT)

SQL: Structured query language (SQL) is a programming language for interacting with a database.